KAIZEN MATHS

Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

S1

In triangle ABC, A = 52°, B = 38°, and a = 10 cm. Find b to 2 d.p.

S2

In triangle ABC, A = 31°, a = 9 cm, and b = 11.46 cm. Find B to 1 d.p.

S3

In triangle ABC, a = 11 cm, b = 7 cm, and included angle C = 76°. Find the area to 2 d.p.

S4

In triangle ABC, A = 66°, B = 64°, and a = 9 cm. Find b to 2 d.p.

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ALG-2B2B-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

E1

In triangle ABC, A = 63°, a = 11 cm, and b = 11.36 cm. Find B to 1 d.p.

E2

In triangle ABC, a = 10 cm, b = 9 cm, and included angle C = 43°. Find the area to 2 d.p.

E3

In triangle ABC, A = 49°, B = 60°, and a = 9 cm. Find b to 2 d.p.

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ALG-2B2B-L1 · E

Show your working

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

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S1

In triangle ABC, A = 52°, B = 38°, and a = 10 cm. Find b to 2 d.p.

Answer:

$b = 7.81$ cm

Working:

Sine Rule: a / sin A = b / sin B

$b = (\sin B / \sin A) \times a$

$b = (\sin 38° / \sin 52°) \times 10$

$\sin 38° \approx 0.6157$, $\sin 52° \approx 0.7880$

$b \approx 7.8129$

$b = 7.81$ cm

S2

In triangle ABC, A = 31°, a = 9 cm, and b = 11.46 cm. Find B to 1 d.p.

Answer:

$B = 41.0°$

Working:

Sine Rule: a / sin A = b / sin B

$\sin B = (b \times \sin A) / a$

$\sin B = (11.46 \times \sin 31°) / 9$

$\sin B \approx 0.6561$

$B \approx \sin^{-1}(0.6561) \approx 41.0°$

$B = 41.0°$

S3

In triangle ABC, a = 11 cm, b = 7 cm, and included angle C = 76°. Find the area to 2 d.p.

Answer:

Area = 37.36 cm²

Working:

Area = 1/2 ab sin C

Area = 1/2 × 11 × 7 × sin 76°

$\sin 76° \approx 0.9703$

Area ≈ 37.3564

Area = 37.36 cm²

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

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S4

In triangle ABC, A = 66°, B = 64°, and a = 9 cm. Find b to 2 d.p.

Answer:

$b = 8.85$ cm

Working:

Sine Rule: a / sin A = b / sin B

$b = (\sin B / \sin A) \times a$

$b = (\sin 64° / \sin 66°) \times 9$

$\sin 64° \approx 0.8988$, $\sin 66° \approx 0.9135$

$b \approx 8.8547$

$b = 8.85$ cm

E1

In triangle ABC, A = 63°, a = 11 cm, and b = 11.36 cm. Find B to 1 d.p.

Answer:

$B = 67.0°$

Working:

Sine Rule: a / sin A = b / sin B

$\sin B = (b \times \sin A) / a$

$\sin B = (11.36 \times \sin 63°) / 11$

$\sin B \approx 0.9205$

$B \approx \sin^{-1}(0.9205) \approx 67.0°$

$B = 67.0°$

E2

In triangle ABC, a = 10 cm, b = 9 cm, and included angle C = 43°. Find the area to 2 d.p.

Answer:

Area = 30.69 cm²

Working:

Area = 1/2 ab sin C

Area = 1/2 × 10 × 9 × sin 43°

$\sin 43° \approx 0.6820$

Area ≈ 30.6899

Area = 30.69 cm²

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

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E3

In triangle ABC, A = 49°, B = 60°, and a = 9 cm. Find b to 2 d.p.

Answer:

$b = 10.33$ cm

Working:

Sine Rule: a / sin A = b / sin B

$b = (\sin B / \sin A) \times a$

$b = (\sin 60° / \sin 49°) \times 9$

$\sin 60° \approx 0.8660$, $\sin 49° \approx 0.7547$

$b \approx 10.3275$

$b = 10.33$ cm